

ADITYA GANDHAMAL

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RESEARCH AND PROFESSIONAL EXPERIENCE

Research Fellow

Sarvam AI

Bangalore

June 2025 – present

- Conducted large-scale data generation for **pre-training** and **SFT** stages of **Sarvam Vision** (3B-parameter VLM, 22 Indian languages) at the scale of **150TB+** and **260M+ images**; sourced data across web, open-source repositories, and built synthetic data generation pipelines, with a focus on Indic OCR, handwritten text, and multilingual document intelligence; model achieved state-of-the-art on olmOCR-Bench and OmniDocBench, **outperforming Gemini 3 Pro, Mistral OCR 3 and GPT 5.2**
- Ideated and designed **data sourcing and labelling strategies** leveraging frontier foundation models, converted the curated datasets from their raw formats to **WebDataset (WebDS)** format for **Nvidia NeMo Framework**-trainable pipelines; currently scaling data sourcing pipelines from **YouTube** and designing data-processing techniques to extract frames-of-interest from YouTube videos to maximise information content and minimise noise.
- Own and maintain the Indic **evaluation and benchmarking pipeline**, designed evaluation strategies for internal Indic benchmarks with OCR-specific metrics including CER, WER, Edit Distance, catastrophic failure counts, and unit-test-style penalisation schemes inspired by RL reward functions; integrated frontier and open-source VLMs (Qwen, Gemma, InternVL, PaddleOCR-VL) with **vLLM-based distributed inference** for scalable evaluation across internal and public benchmarks.
- Leading applied research project for **GoI (MoSPI)**, building a geospatial pipeline integrating **GIS and Satellite data, Gemma4-31B, Segment Any Change** (NeurIPS '26), and **Grounding DINO** to help field officers prioritize regions for **Urban Frame Survey**; pipeline derives decisions with explicit reasoning, assisting in efficiently allocating manual resources for national census; collaborating with GTM team and Government officials to align methodology to operational requirements.

Pre-Doctoral Researcher

Artificial Intelligence and Robotics Laboratory - Indian Institute of Science

Bangalore

Aug 2024 – May 2025

- Conducted research addressing **open-vocabulary learning**, multi-modal alignment, and domain generalization in **VLMs** under the guidance of **Dr. Suresh Sundaram**; core focus on adapting **CLIP** for **open-vocabulary semantic segmentation across unseen categories and new domains**.
- Developed cost aggregation architectures using **Optimal Transport** and **Masked Image Modelling** with parameter-efficient fine-tuning (PeFT) and Prompt-Learning techniques, including **LoRA, Context Optimization, and Conditional Context Optimization** to generalize CLIP across multiple domains for segmentation tasks; explored toggle-based PeFT strategies to address training saturation and improve robustness during inference in open-vocabulary settings.
- Built multimodal benchmarks evaluating segmentation models across varying viewpoints, modalities (**EO-IR/Sim-2-Real**), aerial-to-ground perspectives (**ground vehicles, drone imagery, BEV**) in open-world settings; contributed to data sourcing and formatting, built data ingestion pipelines for ready-to-use evaluation suites, and designed evaluation strategies with uncertainty-based metrics for robustness analysis.
- Developed **CLIPcam**, an open-source Python package (PyPI) for visualizing prompt-image feature alignment in ViT-based CLIP models; delivered a technical talk on **3D Vision** at a Faculty Development Program; contributed to **DRDO-funded** projects; research output spans *2 papers accepted at CVPR Workshops 2025, 1 under review at IROS 2026, and 2 journal papers under review at Transactions on Machine Learning Research (TMLR) and IEEE Transactions on Artificial Intelligence (TAI)*.

Deep Learning and Image Processing Intern

GalaxEye Space

Bangalore

April 2024 – July 2024

- Worked on **SAR-Optical** coregistration and data fusion techniques for GalaxEye's drone-based SAR platform; performed failure-case analysis on drone-captured aerial SAR imagery using **Lucas-Kanade Optical Flow** to study feature distortions and misalignment characteristics; experimented on **SpaceNet-6 with data augmentations to simulate** distorted SAR imagery and developed an **image-warping autoencoder architecture** to model and correct distortions.
- Contributed to the **GLX-SQ payload aboard ISRO's POEM-4 mission** by implementing a coregistration algorithm using **Mutual Information with Powell Optimization**, subsequently porting the Python implementation to C++ for deployment on a Jetson edge device; implemented code profiling to optimize processing for edge deployment, achieving a **~62% reduction in runtime** (40s → 15s).
- Designed a **proof-of-concept SAR-EO fusion architecture** on Spacenet6, exploring **Knowledge Distillation** (leveraging a pre-trained EO encoder) and **Contrastive Learning** (generic cross-modal representation) as complementary fusion strategies, in support of the broader OptoSAR satellite launch — Mission Drishti.

Computer Vision Intern - AI for Healthcare

Indian Institute of Science

Bangalore

Nov 2023 – Feb 2024

- Worked towards the automation of drug delivery systems (Artificial Pancreas) under the guidance of **Dr. Radhakant Padhi**.
- Developed carbohydrate estimation techniques from food images by implementing a meal-detection pipeline integrating **SAM (Segment Anything Model) with ResNet50 and a fuzzy-matching algorithm**, achieving **91% accuracy**.

- Presented findings to a panel of doctors from **Ramaiah Hospital** and **Madras Diabetes Research Foundation**.

Open Source Developer - Machine Learning

Remote

Unify (YC W23)

March 2023 – Oct 2023

- Contributed to **Ivy**, Unify’s open-source framework for cross-framework machine learning interoperability and code transpilation.
- Created and reviewed pull requests involving functional API implementations, framework-specific testing, **code transpilation-related utilities**, bug fixes, and documentation updates across frameworks including **PyTorch, TensorFlow, JAX, and NumPy**; contributed to improving framework consistency, API coverage, and code reliability within Ivy’s unified ML abstraction stack.
- Recognized as a **Top Contributor** within the organization.

Research Intern - NLP

Remote

National Institute of Technology, Kurukshetra

March 2022 – Aug 2022

- Worked on **Link Prediction** on legal documents under the supervision of **Dr. Sarika Jain**; framing it as two sub-tasks: Citation Prediction (predicting which cases a ruling will cite) and Case Similarity (identifying analogous cases sharing legal context).
- Constructed a Legal **Knowledge Graph** from Indian Supreme Court filings using Named Entity Recognition (NER), and benchmarked three GNN architectures (**GraphSAGE, GAT, GCN**) against the baseline RGCN model on both sub-tasks.
- GCN achieved the best results, improving the baseline RGCN AUC by **~40%** on Case Similarity and **~10%** on Citation Prediction.

Project Intern - Computer Vision

Pune

Indian Army

March 2021 – Aug 2021

- Devised techniques for satellite imagery feature extraction, focusing on road network identification for urban planning and border surveillance, under the mentorship of **Col. (Dr.) K. Joshil Raj**.
- **Curated satellite imagery datasets** with their annotations and developed models for **semantic segmentation** (U-NET with ResNet34 as the backbone), fine-tuned to achieve IoU scores of **0.78** and **0.73** for segmenting urban and border roads respectively.
- Developed an interactive GUI with an end-to-end pipeline, allowing model selection and experimentation for Army personnel.

PUBLICATIONS

- **Aditya Gandhamal***, Aniruddh Sikdar*, Suresh Sundaram. “**OV-COAST: Cost Aggregation with Optimal Transport for Open-Vocabulary Semantic Segmentation**” [T4V Workshop, CVPR 2025](#) | [\[paper\]](#) | [\[code\]](#) | [\[poster\]](#)
- Aniruddh Sikdar*, **Aditya Gandhamal***, Suresh Sundaram. “**AetherVision-Bench: An Open-Vocabulary RGB-Infrared Benchmark for Multi-Angle Segmentation across Aerial and Ground Perspectives**” [FMEA Workshop, CVPR 2025](#) | [\[paper\]](#)
- Aniruddh Sikdar*, **Aditya Gandhamal***, Suresh Sundaram. “**CoLoRA: Optimal Transport Meets LoRA for Multi-Domain Open-Vocabulary Segmentation**” *Under Review, IROS 2026*
- Aniruddh Sikdar*, **Aditya Gandhamal***, Suresh Sundaram. “**Tune CLIP as you Toggle (T-CAT): A Novel Toggle-Tune Based Prompt Learning Strategy for Open-Vocabulary Semantic Segmentation**” *Under Review, TMLR*
- **Aditya Gandhamal***, Aniruddh Sikdar*, Arya Kishor, Suresh Sundaram. “**Pixel2Perspective: A Benchmark for Multi-Modal and Multi-View Generalization in Vision-Language Models**” *Under Review, IEEE Transactions on Artificial Intelligence*
- Manjunath D., Prajwal Gurunath, Sumanth Udupa, **Aditya Gandhamal**, Shrikar Madhu, Aniruddh Sikdar, Suresh Sundaram. “**IndraEye: Infrared Electro-Optical UAV-based Perception Dataset for Robust Downstream Tasks**” [arXiv preprint](#)
- **Aditya Gandhamal**, Aniruddh Sikdar. “**CLIPcam: A python package for visualizing prompt-image feature matching in ViT-based CLIP models**” [pip install clip-cam \(PyPI\)](#)
- Sarvam AI - Vision Models Team. “**Sarvam Vision: State-of-the-Art Vision-Language Model for Indic Document Intelligence**” [Technical Blog](#)

EDUCATION

Savitribai Phule Pune University

Aug 2019 – June 2023

Bachelor of Engineering in Electronics and Telecommunication

- Relevant Coursework: Signal Processing, Control Systems, Digital Communication, Data Structures, Object Oriented Programming, Cloud Computing, Database Management Systems, Data Mining

PROJECTS AND OPEN-SOURCE CONTRIBUTIONS

CLIPcam [pip install clip-cam](#)



- Developed **CLIPcam**, an open-source Python package for **visualizing prompt-image** feature matching in ViT-based CLIP models; supports custom fine-tuned checkpoints, **multiple CLIP backbones**, and easy integration into VLM workflows for explainability.

Prompt Learning with CLIP for Open Vocabulary Segmentation



- Implemented prompt learning techniques for **CAT-Seg (CVPR'24)**, integrating CLIP with **Context Optimization (IJCV'22)**, **Conditional Context Optimization (CVPR'22)**, and **Textual-based Class-aware Prompting (CVPR'24)**; conducted ablation studies evaluating each variant against the **MESS benchmark** covering **20 multi-domain** datasets.

Bias Mitigation for Fair Facial Attribute Classification (FAIR-Face)



- Fine-tuned ResNet-50 on CelebA for facial attribute classification while **mitigating spurious correlations in “Bald” and “Male”** attribute predictions; implemented **adversarial training to reduce bias by 20%**, evaluated using Equalized Odds.

Satellite Orbital Trajectory Forecasting (SatPO)



- Developed an LSTM-based time-series forecasting model trained on 18 days of telemetry data to **predict satellite orbital trajectories** for the next 7 days; validated performance through comparative analysis and trajectory visualization.

Ivy — Unified ML Framework

Unify (YC W23) | Top Contributor



- Contributed to Ivy, an open-source framework unifying PyTorch, TensorFlow, and PaddlePaddle for cross-framework code transpilation; **added transpilation features, resolved bugs, implemented and tested functional APIs, and maintained documentation** across **10+ merged pull requests**.

PyTorch-RL



- Added normalisation methods for reward function scaling in **RewardScaling [PR]**; fixed redundant use of `step_mdp` in the data collector, **rectifying incorrect state transitions during rollout [PR]**.

PyTorch-Vision



- Resolved a core **validation bug** where invalid strings were silently accepted as loss reduction modes; implemented **ValueError** handling across relevant loss functions and updated associated tests. [\[PR\]](#)

SKILLS

Languages: Python, Bash/Shell

Vision-Language Models: CLIP, Qwen, Gemma, PaliGemma, InternVL, PaddleOCR-VL

Training Infra & Fine-Tuning: Nvidia DGX Cloud Lepton, Yotta, GCP Training Cluster, LoRA, Prompt Learning, Masked Image Modeling, Knowledge Distillation, Contrastive Learning, Context Optimization

Inference & Deployment: vLLM, Docker, SLURM

Data & Pipelines: WebDataset (WebDS), Data Labelling, Pre-Training Data Curation, Synthetic Data Generation, YouTube Data API v3, Google Cloud Storage

Multi-Modal Sensor Data: RGB-SAR, RGB-IR, Multispectral Satellite Imagery (Cartosat)

Frameworks & Tools: PyTorch, HuggingFace Transformers, Detectron2, vLLM, TensorFlow, OpenCV, Git

TALKS

LiDAR/Camera Fusion for 3D Object Detection | [\[slides\]](#)

Bangalore

Indian Institute of Science – Faculty Development Workshop on AI-Driven Autonomous Drones

Feb 11, 2025

- Covered 3D data representation techniques (**point-based, voxel-based, pillar-based, frustum-based, BEV**) and delivered a literature review spanning **LiDAR-based methods (PointNet++, VoxelNet, PointPillars, SECOND)**, **camera-based methods (FCOS3D, PETR, DETR3D)**, and **multimodal fusion architectures (MVX-Net, BEVFusion)**.
- Discussed real-world applications in autonomous driving and robotics, key challenges including sensor data sparsity and adverse weather robustness, and standard benchmarks — **KITTI, Waymo, and nuScenes**.

ACHIEVEMENTS

- 2x Awardee of **KIAC Fellowship (Kotak-IISc AI-ML Centre)**
- Algorithm developed during an internship at **GalaxyEye Space** was successfully deployed and operational aboard the **GLX-SQ payload** launched as part of **ISRO's POEM-4 Mission**.
- Contributed to building **Sarvam Vision**, India's first **sovereign Vision-Language Model**, achieving **state-of-the-art** performance on Indic OCR and global document understanding benchmarks.

VOLUNTEERING

AI/ML Co-Lead

Pune

Google Developer Student Clubs

Aug 2021 – July 2022

- Organized and led **Google Developer technology events**, including hackathons and workshops. Conducted hands-on sessions on **“Tensorflow”** and **“Git & GitHub”**, reaching an attendance of 150 students overall.